FNC = „formă normală conjunctivă"

= grupări de $\vee$ legate de $\wedge$-uri

e.g $(a \vee b) \wedge (b \vee c \vee \alpha) \wedge (a \vee \alpha) \wedge \ldots$

$\sim$ cu intrările semn tab. de adevăr

Dacă am o <u>mult.</u> de clauze din FNC pot să determin dacă e satisfiabilă cu alg. <u>Davis - Putnam</u>

Voi scrie $a \vee b$ ca $\{a, b\}$.

1 pas Davis - Putnam = elim. unei variabile

ex. $\Sigma_0 = \{\{\alpha, \neg\beta\}, \{\alpha, \delta, \varepsilon\}, \{\neg\alpha, \neg\delta\}, \{\neg\alpha, \gamma\}, \{\beta, \gamma\}\}$

vreau să elimin pe <u>α</u>:

împart în:

→ expresii cu α: $\boxed{\{\alpha, \neg\beta\}, \{\alpha, \delta, \varepsilon\}}$

→ expresii cu $\neg\alpha$: $\boxed{\{\neg\alpha, \neg\delta\}, \{\neg\alpha, \gamma\}}$

→ restul (nu contează)

$\{\beta, \gamma\}$

• → scot ce avea cu α => $\{\{\neg\beta\}, \{\delta, \varepsilon\}\}$

• → scot tot $\neg\alpha$ => $\{\{\neg\delta\} \{\gamma\}\}$

fac toate combinațiile

1 element •, 1 element •

Obțin $\boxed{\{\neg\beta, \neg\delta\}}, \boxed{\{\neg\beta, \gamma\}}, \underbrace{\{\delta, \varepsilon, \neg\delta\}}_{\Downarrow}, \boxed{\{\delta, \varepsilon, \gamma\}}$

însemna $\delta \vee \varepsilon \vee \neg\delta$

mereu adev. => <u>îl scot</u>

$\boxed{\text{ce rămâne}}$ => adaug peste restul

$\Sigma_0 \to \Sigma_1$:

$$\Sigma_1 = \{\underbrace{\{\beta, \gamma\}}_{\text{inițial; nu avea pe } \alpha}, \underbrace{\{\neg\beta, \neg\delta\}, \{\neg\beta, \gamma\}, \{\delta, \varepsilon, \gamma\}}_{\text{am adăg. mai sus}}\}$$

<u>Obs.</u> Pt. programe e nevoie, fără explicații combinatoriale (dar pt. ex. la examen e util)

DEM. DAVIS - PUTNAM, EX. CMC 2023, EX. 14.1.b)

$$\Sigma = \{ \underset{1)}{\neg\alpha \to (\beta \wedge \gamma)}, \ \underset{2)}{\alpha \longleftrightarrow (\beta \vee \gamma)} \}$$

Nu au din prima FNC => transform!

D.R. $\boxed{\alpha \to \beta \sim \neg\alpha \vee \beta, \quad \alpha \Leftrightarrow \beta \sim \alpha \to \beta \wedge \beta \to \alpha}$

1) $\neg\alpha \to (\beta \wedge \gamma) \sim \neg\neg\alpha \vee (\beta \wedge \gamma) \sim \alpha \vee (\beta \wedge \gamma) \sim$

$\sim (\alpha \vee \beta) \wedge (\alpha \vee \gamma) \Rightarrow \underline{\{\alpha, \beta\}, \{\alpha, \gamma\}}$

2.1)

2) $\alpha \longleftrightarrow (\beta \vee \gamma) \sim \boxed{\alpha \to \beta \vee \gamma}$ și $\boxed{\beta \vee \gamma \to \alpha}$

2.2)

2.1) $\alpha \to \beta \vee \gamma \sim \neg\alpha \vee (\beta \vee \gamma) \sim \neg\alpha \vee \beta \vee \gamma$

$\Rightarrow \underline{\{\neg\alpha, \beta, \gamma\}}$

2.2) $\beta \vee \gamma \to \alpha \sim \neg(\beta \vee \gamma) \vee \alpha \sim (\neg\beta \wedge \neg\gamma) \vee \alpha$

$\sim (\neg\beta \vee \alpha) \wedge (\neg\gamma \vee \alpha) \Rightarrow \underline{\{\neg\beta, \alpha\}, \{\neg\gamma, \alpha\}}$

$$\Sigma = \{ \overset{c_1}{\{\alpha, \beta\}}, \overset{c_2}{\{\alpha, \gamma\}}, \overset{c_3}{\{\neg\alpha, \beta, \gamma\}}, \overset{c_4}{\{\neg\beta, \alpha\}}, \overset{c_5}{\{\neg\gamma, \alpha\}} \}$$

clauze cu α: c_1, c_2, c_4, c_5, cu $\neg\alpha$: c_3.

$\downarrow$ elimin α $\qquad$ $\downarrow$ elimin $\neg\alpha$

$\{\beta\}, \{\gamma\}, \{\neg\beta, \bullet\}, \{\neg\gamma\} \quad \times \quad \{\beta, \gamma\}$

$\{\beta, \beta, \gamma\}, \{\beta, \gamma, \gamma\}, \underbrace{\{\neg\beta, \beta, \gamma\}}_{1}, \underbrace{\{\neg\gamma, \beta, \gamma\}}_{1}$